

MoiréSkin: Ultra-Sensitive Sensor–Actuator Visuo-Tactile Skin Using Moiré Patterns

Paper #1177, 12 Pages

ABSTRACT

Robots are increasingly expected to perform dexterous object manipulation in real-world environments. However, existing visuo-tactile sensors (VTS), despite their rich camera-based spatial readout, largely rely on rigid optical structures that limit contact force sensitivity, measurable force range, and active compliance regulation. We present *MoiréSkin*, an inflatable visuo-tactile skin that uses flexible Moiré optics to unify ultra-sensitive tactile sensing and active compliance regulation. It embeds dual-sided sub-millimeter gratings in a soft membrane, transforming microscopic contact-induced strain into large, high-contrast Moiré fringe variations for amplified tactile readout. To balance reliability and sensing sensitivity, we develop a tailored soft-material grating sensor fabrication pipeline, integrate internal illumination with fluorescent doping to suppress ambient-light interference. A self-supervised calibration module is designed to compensate for air-pressure-dependent sensing drift during pneumatic regulation. We further propose a neural inference pipeline that takes Moiré fringes before and after contact as input to estimate contact position and force. Experiments show that *MoiréSkin* achieves 0.094 mm position and 0.0038 N contact force MAE, outperforming prior VTS by an order of magnitude. It also reaches 99.7% 10-shot classification accuracy over 7 ultra-soft materials spanning 8.73–21.72 kPa Young’s modulus, and measures shear-force-related liquid mass variation with ~ 4 g precision, demonstrating a new path toward ultra-sensitive sensor-actuator visuo-tactile skins.

1 INTRODUCTION

Robots are increasingly expected to perform everyday dexterous object manipulation in real-world environments, e.g., picking up fragile groceries, opening containers, inserting connectors, and handling deformable packaging. Yet these tasks are ultimately governed by *contact*: forces depend on geometry and frictional surface properties, contact patches evolve during motion, and brief transients can turn a seemingly correct grasp into slip or damage. Achieving reliable dexterity therefore requires closing the loop at the interface: robots must not only *sense* contact forces and slip risk, but also actively *shape* contact conditions in real time.

Human fingertips illustrate the level of precision and reactivity required. First, *dense tactile innervation* provides rich, localized feedback: a single fingertip is innervated by roughly 10^3 low-threshold tactile afferents, with mechanoreceptive unit densities reaching ~ 241 units/cm² [1, 2]. Second, *fine-grained force sensing* supports manipulation in the low-Newton regime: humans can scale and discriminate forces typical of manipulation (~ 0.1 –5 N) [3], where changes on the order of *tens of mN* can be behaviorally meaningful for

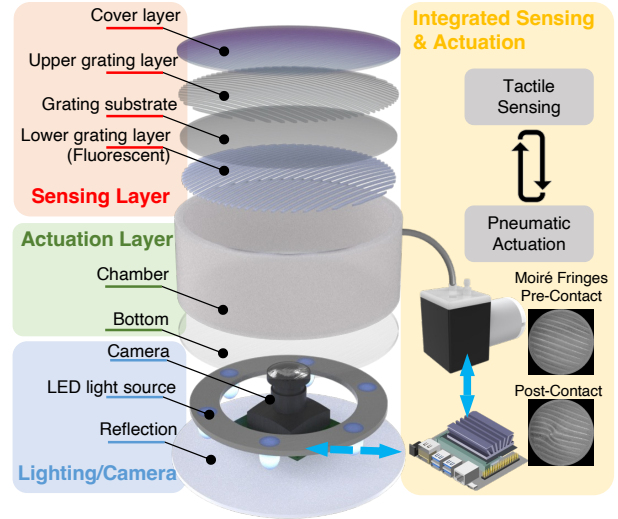


Figure 1: *MoiréSkin* overall architecture. It embeds Moiré patterns into a flexible film to enable ultra-sensitive integrated visuo-tactile sensing and actuation.

grasp stability. Third, *active compliance regulation* stabilizes grasps under friction uncertainty: during precision grip, humans maintain grip force only slightly above the slip threshold rather than relying on conservative over-squeezing [4, 5]. Together, these properties motivate a clear design target for robotic tactile skin: localized, high-sensitivity sensing paired with contact-aware actuation.

Traditional resistive/capacitive tactile sensor arrays are difficult to scale to dense, fine-grained tactile skins as higher density increases wiring/packaging complexity, reduces conformability, and exacerbates shear- and wear-induced drift [6]. Recent Visuo-Tactile Sensing (VTS) instead uses cameras to infer contact properties from surface deformation. However, existing VTS designs suffer from a fundamental *sensitivity-compliance paradox*. Rigid VTS offers high spatial resolution but relies on stiff elastomers that saturate under large loads and lack the compliance needed for safe, irregular contact [7, 8]. Conversely, non-rigid VTS adopts compliant, air-filled interfaces but is bottlenecked by low spatial resolution, often remaining at sub-Newton precision due to the limits of sparse marker-tracking [9]. Furthermore, both classes decouple tactile sensing from active compliance regulation, limiting integrated sensor-actuator capabilities for dexterous manipulation. Tab. 1 summarizes their limitations.

More specifically, *MoiréSkin* targets contact regimes in which four capabilities must coexist: tiny-force sensitivity, non-rigid air-inflated compliance, integrated sensing–actuation, and see-through visual access before contact. This combination is useful for gram-scale loads, fragile or deformable contacts, changing loads that call for pressure regulation, and

interactions that benefit from observing the object immediately before contact. Tab. 1, Fig. 13, and the studies in Sec. 7.3 demonstrate these advantages on the current prototype.

To address these, we propose *MoiréSkin*, a novel non-rigid inflatable visuo-tactile skin based on Moiré-based sensing. Our key idea is to embed *dual-sided, sub-millimeter optical gratings* into a flexible film to generate *Moiré fringes*, low-frequency beat patterns arising from microscopic geometric mismatches. These fringes provide massive geometric amplification, magnifying tiny relative grating displacements into visible Moiré fringe distortions during external contact or internal pneumatic actuation. By decoding these distortions, *MoiréSkin* estimates both contact force and actuation state, enabling closed-loop control that can simultaneously *measure* and *shape* contact during manipulation.

MoiréSkin faces four technical challenges.

How to balance sensor sensitivity with fabrication reliability? *MoiréSkin* relies on sub-millimeter optical gratings whose pitch and contrast must remain uniform after transfer onto a compliant substrate. Moreover, the two grating layers must be precisely aligned to generate stable Moiré fringes. We design a precise simulation tool to characterize the key fabrication parameters and guide the design space. Based on these insights, we establish a fabrication pipeline that preserves grating fidelity and alignment at scale while maintaining mechanical durability. We further tailor the soft material properties to achieve a favorable balance among compliance, deformability, stretchability, and optical transparency.

How to make MoiréSkin robust under ambient lighting conditions without losing “see-through” functionality? Existing structured-illumination-based VTS typically use an opaque cover layer to shield the sensing region from ambient light interference, which prevents the camera from observing the object before contact. In contrast, *MoiréSkin* aims to preserve “see-through” capability by using a transparent cover layer. However, this design introduces the challenge of ensuring that the observed Moiré fringes are not corrupted by the appearance of the contact object. We develop a fluorescence-assisted fabrication process in which one high-frequency grating excites fluorescent emission, thereby enhancing the contrast of the generated Moiré fringes under diverse ambient lighting conditions. In addition, we co-design the internal illumination to produce a uniform light field while minimizing glare and lens flare.

How to sense and calibrate the actuation state across different atmospheric pressure? *MoiréSkin* uses pneumatic actuation to enable closed-loop control. Without accurate real-time estimation of the actuation state, such a control loop cannot reliably achieve the desired deformation. However, accurate state estimation is challenging because both material properties and atmospheric pressure significantly affect the actuation response. To address it, we first perform a one-time material calibration by learning a Moiré-fringe-to-actuation-state conversion model. We then propose a self-supervised

calibration algorithm that automatically identifies the zero-actuation state, i.e., when the internal chamber pressure equals the external atmospheric pressure.

How to achieve accurate contact position and force sensing across different actuation states? The Moiré fringe response depends not only on the contact itself, but also on the instantaneous actuation state of *MoiréSkin*. To address this challenge, we design a neural network framework that captures local Moiré fringe variations by taking the pre-contact and post-contact fringe patterns as input. A two-stage model is designed to refine tactile sensing results for higher spatial resolution and force estimation accuracy. Finally, we introduce a series of data augmentation strategies to improve model robustness across diverse sensing conditions.

We implement an end-to-end hardware/ software prototype of *MoiréSkin*, as shown in Fig. 1. The hardware architecture consists of three components: (1) a sensing layer that generates Moiré fringes, (2) an actuation layer formed by a sealed chamber connected to a pneumatic actuator, and (3) a lighting and imaging layer comprising an RGB camera and LED illumination. Both the pneumatic actuator and the camera are interfaced with a shared control and computation unit, which captures the Moiré fringes and enables integrated sensing and actuation. To evaluate the system, we collect a dataset of more than 20,000 samples spanning diverse contact forces and positions. We further conduct two case studies: (1) ultra-soft material stiffness classification and (2) mass estimation via shear force, to demonstrate the benefits of integrated sensing and actuation. Extensive experiments show that *MoiréSkin* achieves 0.094 mm position and 0.0038 N contact force MAE. It also achieves 99.7% 10-shot classification accuracy across seven ultra-soft materials with Young’s moduli ranging from 8.73 to 21.72 kPa, and measures shear-force-induced liquid mass variation with 4 g precision.

Our contributions are summarized as follows:

- We introduce *MoiréSkin*, the first inflated VTS system to leverage stretchable dual-sided high-frequency gratings for generating ultra-sensitive Moiré fringes.
- We develop a specialized soft-material fabrication pipeline to ensure high resolution and ambient-light suppression.
- We propose a physics-informed framework that calibrates material properties and compensates for atmospheric-pressure variation, along with a two-stage neural inference pipeline for simultaneous contact position and force estimation.
- Extensive experiments show that *MoiréSkin* achieves SoTA tactile sensing performance, outperforming prior visuo-tactile sensing systems by an order of magnitude, while enabling integrated sensing and actuation.

2 RELATED WORKS

Robotic Tactile Sensing. Early robotic tactile systems mainly relied on discrete point-contact arrays using piezoresistive or capacitive transduction [10, 11]. Although effective for basic contact detection, these architectures suffered from complex wiring and limited ability to capture continuous and accurate deformation or shear-force distributions [12]. Vision-based

Table 1: Benchmarking MoiréSkin against SOTA VTS (F.: Normal Force; P.: Position; Sense & Act: Integrated Sensing & Actuation Ability)

Sensor	F. MAE (N)	P. MAE (mm)	Compliance	Sense & Act
[24]	—	—	Elastic skin on rigid frame	×
[25]	—	—	Active mobile surface	✓
[26]	—	$0.4_{<0.5}$	Rigid core with cast gel	×
[7]	$1.9_{\geq 0.1}$	$0.03_{<0.1}$	Rigid base w/ soft elastomer	×
[9]	$0.8_{\geq 0.1}$	—	Non-rigid (Air-inflated)	×
[27]	$0.41_{\geq 0.1}$	$0.3633_{<0.5}$	Elastomer on rigid base	×
[28, 29]	$0.307_{\geq 0.1}$	$0.0462_{<0.1}$	Soft silicone on rigid base	×
[30]	$0.25_{\geq 0.1}$	—	Gratings on rigid substrate	×
[8]	$0.04_{<0.1}$	$0.96_{\geq 0.5}$	Rigid finger base	×
[31]	$0.03_{<0.1}$	$0.4_{<0.5}$	Single elastomer on cone	×
[32, 33]	$0.01_{<0.1}$	$0.2_{<0.5}$	Non-rigid (3D dome)	×
ours	$0.0038_{<0.01}$	$0.094_{<0.1}$	Non-rigid (Air-inflated)	✓

tactile sensors addressed these limitations by placing cameras behind deformable sensing layers and recasting tactile measurement as an image-based inference problem [13–17]. GelSight, for example, demonstrated high-resolution tactile sensing based on 3D surface reconstruction via photometric stereo [7], while later systems extended the design toward dense force sensing, slip detection, and compact embodiments [18–23]. MoiréSkin preserves the benefits of high-dimensional VTS but departs from their reliance on complex structured illumination design. Instead, it uses a thin inflatable membrane with Moiré fringes to jointly enable ultra-sensitive optical tactile sensing and active compliance regulation.

Sensitivity vs. Compliance. Translating contact-induced deformation into visual cues introduces a fundamental trade-off between sensitivity and compliance [34]. Optical intensity-mapping mechanisms that achieve the highest spatial resolution rely on stiff substrates to avoid out-of-focus visual cues as well as unstable or chaotic optical distortions [7, 27, 29, 35–40]. However, such stiffness reduces physical compliance and limits the sensor’s ability to safely conform to irregular objects. Alternatively, tracking the kinematic movement of discrete fiducial markers allows systems to utilize softer substrates, but their spatial resolution is strictly bottlenecked by marker density [24, 32, 33, 41–43]. MoiréSkin fundamentally addresses this sensitivity-compliance trade-off by decoupling the Moiré-based optical amplification mechanism from the mechanical stiffness of the contact medium. As a result, MoiréSkin achieves both extreme volumetric compliance and ultra-high tactile sensitivity while enabling integrated sensing and actuation, as shown in Tab. 1.

Moiré-Based Sensing.

Prior Moiré visual sensing falls into several mechanism classes. Stacked rigid gratings support tracking and tangible interaction through markers and widgets [44–46]. Camera-screen or color-filter-array interference instead encodes pose, motion, and vibration [47–49], while mID uses screen-camera Moiré patterns for photo forensics [50]. MoiréEar captures sound-induced motion [51]. MoiréLens images refractive disturbances caused by airflow, heat, and gas [52].

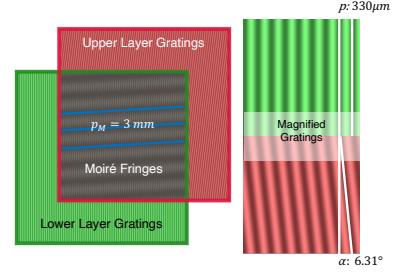


Figure 2: Moiré Fringes Generation Model.

These systems amplify rigid-body motion, screen-camera geometry, vibration, or other non-contact phenomena. Concurrent work, MoiréTac, applies Moiré sensing to a rigid planar surface [30]. In contrast, MoiréSkin embeds opposing gratings in a highly compliant inflatable film, converting contact-induced non-rigid deformation into a tactile signal. The same fringe rotation also feeds pneumatic state control, making Moiré both the tactile observation channel and the feedback signal for active compliance regulation.

3 PRIMER AND PRELIMINARY STUDY

First, we introduce the basic primer underlying MoiréSkin, which uses distortions in Moiré fringes on a soft elastomeric film to transduce contact-induced deformation for tactile sensing. Second, we present simulation-based preliminary studies to characterize the trade-off among spatial sensitivity, deformation sensitivity, and fabrication reliability.

3.1 Moiré Fringes for Tactile Sensing

Moiré Generation. Moiré fringes are low-frequency patterns formed by superimposing two similar high-frequency gratings with a small geometric mismatch [53]. Fig. 2 illustrates the Moiré fringe generation model used in MoiréSkin. Specifically, MoiréSkin patterns identical periodic gratings on the upper and lower layers of a thin, compliant elastomeric film. The grating pitch p denotes distance between adjacent grating lines. When the two gratings are superimposed with a relative angular mismatch α , they produce low-frequency Moiré fringes with pitch

$$p_M = \frac{p}{2 \sin\left(\frac{\alpha}{2}\right)}. \quad (1)$$

When α is small, the Moiré fringes p_M become much larger than the underlying grating pitch p , visually amplifying microscopic geometric mismatch into prominent large-scale fringes. Fig. 2 shows a zoomed-in example: when $p = 330 \mu\text{m}$ and $\alpha = 6.31^\circ$, the resulting Moiré fringe pitch is $p_M \approx 3.0 \text{ mm}$.

Deformation-Induced Moiré Fringes Change. This geometric amplification makes Moiré fringes well suited for tactile sensing. When contact is applied to the compliant film, the membrane deforms locally. Because the gratings are patterned on opposite sides of the film and separated by its thickness, they do not deform identically. Their local strain, relative registration, and effective pitch can therefore diverge, perturbing the local Moiré fringe generation condition. For example,

Fig. 3 shows the simulated Moiré fringe rotation under different levels of global stretching. *MoiréSkin* leverages these amplified Moiré fringe signatures to infer tactile quantities such as contact location and contact normal/shear force.

3.2 Sensitivity Preliminary Study

Next, we conduct simulation-based preliminary studies on key design parameters, i.e., grating pitch p and grating angular mismatch α , to understand the trade-off among spatial sensitivity, deformation sensitivity, and fabrication robustness.

3.2.1 Spatial Sensitivity. *Smaller grating pitch p and larger angular mismatch α improve the spatial sensitivity by producing denser Moiré fringes within the same sensing area.*

Spatial sensitivity characterizes the ability of *MoiréSkin* to localize contact-induced deformation. It is mainly determined by the number of Moiré fringes captured within the camera region of interest (RoI), since denser fringes provide a finer spatial representation of local deformation. For consistent analysis, we define the default RoI as a circular area with diameter of 30 mm ($\sim 707 \text{ mm}^2$). Although our analysis is based on this default scale, the same principles generalize to other form factors. To reduce boundary artifacts, the compliant elastomeric film is designed to be slightly larger than the RoI, with a diameter of 35 mm. This margin mitigates edge effects and minimizes interference from the surrounding rigid frame, which could introduce nonlinear distortion.

Using Eq. 1, we analyze how p and α jointly determine the Moiré fringe pitch p_M . We sweep both parameters and convert the resulting p_M into the number of Moiré fringes within the RoI to visualize spatial sensitivity (Fig. 4 (b)). The results show that smaller p and larger α lead to higher spatial sensitivity. In practice, however, the minimum achievable grating pitch is constrained by our current soft-lithography fabrication pipeline to $330 \mu\text{m}$ (discussed in Sec. 4). We therefore fix $p = 330 \mu\text{m}$ and further visualize the generated Moiré fringes under different α in Fig. 4 (c). *With a fixed p , a higher α gives *MoiréSkin* better spatial sensitivity by producing denser Moiré fringes.*

3.2.2 Deformation Sensitivity. *Deformation sensitivity is primarily governed by the angular mismatch α . Smaller α improves deformation sensitivity by producing larger Moiré fringe distortions under the same mechanical deformation.*

Deformation sensitivity characterizes the ability to resolve mechanical deformation of the elastomeric film. In our system, this sensitivity is primarily reflected by the orientation angle of the Moiré fringes, denoted by θ_M . For simplicity, we first consider global deformation, which causes a global change in θ_M across the RoI. Under localized contact, the same principle applies locally and produces spatially varying θ_M . As shown in Fig. 3, the Moiré fringe orientation θ_M can be modeled as

$$\theta_M = \arctan\left(\frac{p_{\text{upper}} \sin(\alpha)}{p_{\text{upper}} \cos(\alpha) - p_{\text{lower}}}\right), \quad (2)$$

where p_{upper} and p_{lower} denote the grating pitches of the upper and lower layers, respectively. As shown in Fig. 3 (a), Before

deformation, the two pitches are identical, i.e., $p_{\text{upper}}^{(0)} = p_{\text{lower}}^{(0)}$. After deformation, however, depth separation causes the two layers to experience different strain, resulting in $p_{\text{upper}}^{(t)} \neq p_{\text{lower}}^{(t)}$ at time t (Fig. 3 (b), (c)). This pitch mismatch perturbs the Moiré fringe orientation and converts subtle deformation into amplified Moiré fringes variations.

To quantify this effect, we define the deformation sensitivity as $S_d = \frac{d\Delta\theta_M}{dR_p}$, where $R_p = p_{\text{upper}}/p_{\text{lower}}$ is the deformation-induced pitch ratio and $\Delta\theta_M$ is the corresponding fringe orientation change. For a given change in pitch ratio R_p , a larger variation in $\Delta\theta_M$ indicates higher deformation sensitivity. Fig. 4 (a) presents the simulated deformation sensitivity S_d . The results reveal two key insights: (1) *the geometric amplification gain increases as the angular mismatch α approaches zero*; and (2) *the deformation sensitivity is primarily determined by α and is decoupled from the grating pitch p .*

3.2.3 Sensitivity Trade-off and Design Choice. Our preliminary studies reveal a fundamental design trade-off in *MoiréSkin*. With the grating pitch $p = 330 \mu\text{m}$ fixed by fabrication constraints, the angular mismatch α becomes the main parameter for balancing spatial and deformation sensitivity. A larger α improves spatial sensitivity but reduces deformation sensitivity. *MoiréSkin can be made task-adaptive by selecting α during fabrication according to the desired sensing objective. In our prototype, the commercial mold and transfer fidelity fix $p = 330 \mu\text{m}$. We therefore choose $\alpha = 6.31^\circ$, yielding $p_M = 3 \text{ mm}$ (approximately ten fringes across the 30 mm RoI), as a balanced design point that jointly supports spatial localization and tiny-force deformation amplification (Fig. 4(c)).*

4 HARDWARE DESIGN

We next describe the hardware construction of *MoiréSkin*. As shown in Fig. 1, the sensing layer consists of multiple flexible film layers that generate high-contrast Moiré fringes, which amplifies contact-induced deformation for tactile sensing. The actuation layer contains a sealed pneumatic chamber for actuation and active compliance. The lighting/ camera layer integrates a commercial RGB camera with dedicated illumination for stable visual readout. The camera and pneumatic actuator are connected to a shared controller and computation unit, enabling integrated sensing and actuation. This section presents three key hardware components.

4.1 Materials and Fabrication Design

We first design the material stack and fabrication process to form high-fidelity Moiré fringes on a soft substrate.

The fabrication novelty is a simulation-guided soft-material transfer pipeline that registers high-frequency gratings on opposite sides of one compliant film while preserving the optical contrast needed for Moiré amplification.

4.1.1 Mold and Material Preparation. **Preparation of patterned mold.** *We prepare a patterned mold that transfers the grating geometry to the soft layers.* The patterned mold is

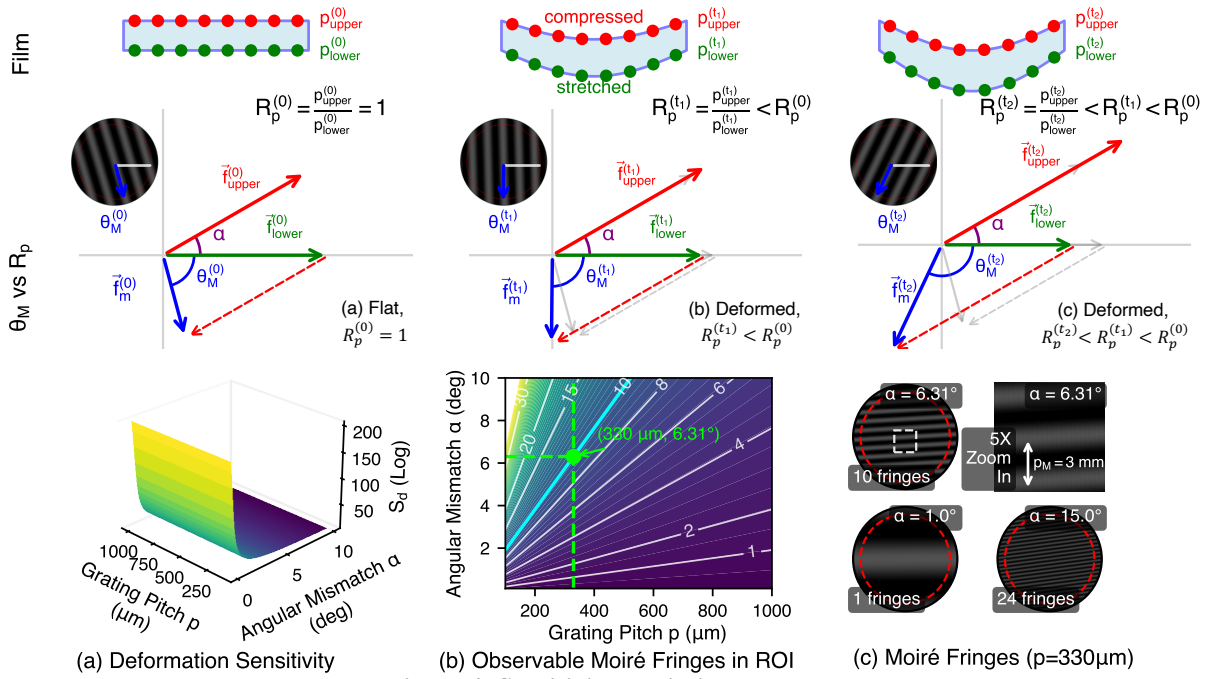


Figure 4: Sensitivity preliminary study.

fabricated via soft lithography, with a uniform grating pitch of $p = 330 \mu\text{m}$. Because transfer fidelity directly affects the resulting Moiré fringes quality, the mold surface must be clean and uniform. We therefore sequentially rinse the custom optical grating mold with acetone, isopropyl alcohol, and deionized water, followed by nitrogen drying. The mold is then plasma treated at 30 W for 2 min to remove residual organic matter, increase surface hydrophilicity, and improve pattern transfer reliability.

Preparation of liquid silicone rubber (LSR). We next prepare the soft LSR used throughout MoiréSkin. The elastomer simultaneously forms the mechanical interface, grating carrier, and optical path. We select it for compliance and deformability under tiny contact forces, stretchability under pneumatic strain, optical transparency, grating-transfer fidelity, and soft-lithography compatibility. Our formulation mixes Silicones XP-565 base, Sylgard 527 softening agent, and curing agent at a 9:1:1 mass ratio.

4.1.2 Dual-side Grating Fabrication Pipeline. To produce Moiré fringes that amplify contact-induced deformation, MoiréSkin patterns high-frequency gratings on dual-sided layers. Fig. 5 illustrates the detailed fabrication pipeline of the dual-sided grating structure in Fig. 1, including the grating substrate and the upper and lower grating layers.

Grating Substrate Fabrication. We first fabricate the grating substrate. The LSR is manually stirred for 5 min until homogeneous, followed by 2 min of ultrasonication. To guarantee the fabrication quality, the mixture is then placed in a vacuum desiccator, and degassed until visible bubbles are removed. Then, the degassed LSR was cast onto a flat mold to form a uniform layer and cured in room temperature to obtain the grating substrate. This casting process enables the fabrication of grating substrates with thicknesses exceeding $100\mu\text{m}$, allowing substrate thickness control by adjusting the casting volume.

Single-Layer Grating Casting. Next, we fabricate a single-layer grating film by casting the degassed grating substrate onto the patterned mold, following a spin coating-imprinting-curing strategy processes (Fig. 5 Step 7-9). At Step 7, a volume of 2–3 mL LSR is dispensed onto cover the grating patterned mold, and spin-coated to form a homogeneous thin film. The coated layer is then subjected to a preliminary low-temperature curing step to bring the material into a semi-cured state, to stabilize the geometry of the replicated grating. At Step 8, the mold with the thin film is flipped so that the patterned surface faces downward and is brought into conformal contact with the previously prepared structure. A microscale imprinting process is then applied by exerting gentle pressure, allowing the semi-cured LSR film to conformally fill the microcavities of the mold and replicate the patterned features. A vacuum treatment is applied to remove trapped bubbles and improve both pattern fidelity and optical quality, then thermally cured in a vacuum oven. At Step 9, The film is then soft-baked at 65°C under vacuum for 10–15 min to stabilize the surface relief, after which the patterned elastomer film is gently peeled off. The resulting film carries a faithful negative replica of the grating pattern while maintaining the flexibility and optical transparency required for Moiré-based sensing.

Double-Layer Grating Casting. A straightforward way to form the second grating would be to repeat the same process on the opposite side with a small angular offset α . Instead, we adopt a modified fabrication design for the bottom grating layer so that it can additionally provide fluorescent reflection, which reduces interference from ambient-light interference. We describe this bottom grating layer fabrication strategy (Step 2-6) in the next section.

Assembly. Finally, the dual-grating films are inspected under an optical microscope to verify structural integrity. Samples with delamination, incomplete pattern transfer, or residual

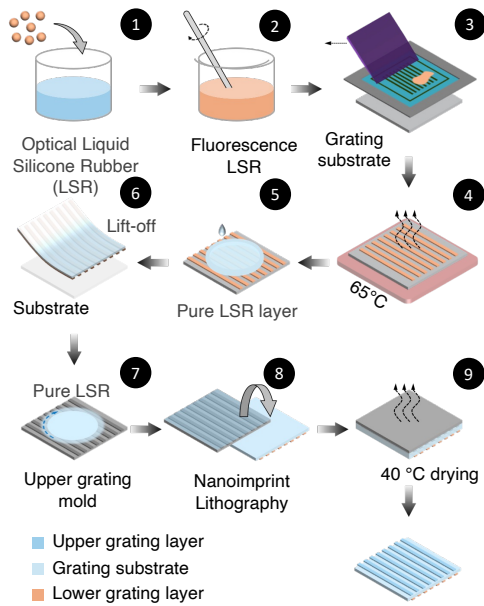


Figure 5: Dual-side grating fabrication pipeline.

bubbles are discarded. Completed films are stored in a dust-free Petri dish at room temperature and remain stable for at least three months without noticeable degradation. Across more than 20,000 tactile readouts collected with the same MoiréSkin cell, the membrane and gratings remained visually intact, and the same calibration supported the full evaluation campaign. This repeated-use result supports prototype robustness and motivates dedicated long-term pneumatic-cycling and shear-fatigue characterization. To complete the MoiréSkin cell, we integrate the dual-sided grating structure with a protective top cover and a sealed pneumatic chamber underneath, as shown in Fig. 1. The top cover is formed using a commercial spray-on TPU elastomer, which provides a thin protective coating while preserving the optical readout and mechanical compliance of the sensor.

4.2 Optical Design

A common challenge for VTS is that ambient light outside the cover layer can corrupt the captured tactile sensing image. To mitigate this challenge, many structured-light-based VTS designs use opaque cover layers [7]. However, doing so blocks the camera’s ability to see through the sensing surface, preventing pre-contact perception and limiting shape observation during the transition from non-contact to contact [54]. MoiréSkin addresses ambient-light interference while preserving “see-through” capability through a high-contrast *fluorescent* Moiré fringes generation and optical co-design.

The optical novelty is the co-design of fluorescent doping, a spectrally selective transparent cover, and controlled internal excitation. The dopant raises grating visibility, the cover suppresses out-of-band ambient interference, and the LED ring provides a repeatable excitation source. Fig. 6 evaluates the resulting integrated stack under varying ambient-light conditions.

4.2.1 Fluorescent Fabrication and Casting. As shown in Fig. 5 Step 2, long-afterglow luminescent materials are

incorporated into LSR to form the lower grating layer. The luminescent powder is firstly dried at 80 °C for 12 h to remove adsorbed moisture. After cooling to room temperature in a desiccator, the powder was mixed with LSR at a mass ratio of 1:20. To ensure uniform dispersion and stable material properties, the powder is gradually added under continuous stirring at low temperature. Step 3-6 involves patterned printing, thermal curing, LSR casting, and transfer printing. First, the fluorescent LSR is transferred onto the substrate by patterned printing, forming the desired grating pattern. The printed structures are then thermally cured at 65 °C to stabilize the printed features and improve their mechanical integrity. Then LSR is cast onto the patterned surface, allowing the LSR to infiltrate and conformally cover the micro-structured features. After curing, the fluorescent patterned structures will be transferred onto the elastomeric film, then the entire layer is peeled off from the substrate for the following process. This process enables the bottom grating layer to emit fluorescent signals that enhance Moiré visibility while reducing sensitivity to external illumination.

4.2.2 Cover Layer Design. The cover layer is co-designed to further enhance the contrast of the fluorescent Moiré fringes. Specifically, it is engineered with spectral filtering properties that attenuate long-wavelength red light (approximately 620–750 nm) from the environment while allowing the desired fluorescent emission to pass. By suppressing unwanted background components, this design improves the visibility of Moiré fringes without sacrificing transparency.

4.2.3 Illumination Co-Design. The illumination system in MoiréSkin serves two purposes: (1) exciting the embedded fluorescent gratings, and (2) providing a stable optical baseline for camera exposure. To this end, MoiréSkin uses a programmable WS2812 LED ring tuned to a blue excitation band (455–485 nm), which matches the excitation spectrum of the fluorescent material embedded in the lower grating layer. As shown in Fig. 1, the LED ring is oriented downward toward a diffuse reflection plate at the base of the module. This configuration converts discrete LED sources into a more uniform internal light field, reducing local hotspots on reflective elastomer surfaces. The LED ring is also mounted below the camera optics so that the surrounding structure acts as a baffle, blocking direct light paths into the lens and suppressing glare and lens flare. Together, these optical components enable MoiréSkin to retain “see-through” functionality while remaining robust to ambient-light interference.

4.3 Actuation Hardware Design

The actuation novelty is to use the same compliant optical membrane for tactile transduction and pressure-regulated compliance, with the camera-derived fringe state closing the pneumatic feedback loop. To tightly couple tactile sensing with active compliance regulation, MoiréSkin integrates a sealed pneumatic chamber beneath the dual-layer grating structure, as shown in Fig. 1. The chamber is connected to an air pump, and both the pump and the camera interface with the

same backend computing unit, enabling closed-loop sensing and actuation. Through this shared backend, MoiréSkin can regulate contact conditions based on real-time tactile feedback. For example, when the sensing module detects incipient slip from shear-induced Moiré changes, the controller can increase chamber pressure to stiffen the contact interface and raise the available friction force. Conversely, the pressure can be reduced to soften the interface for gentler contact, such as when interacting with fragile objects.

Such sensing-actuation integration is difficult to achieve with conventional structured-light-based rigid VTS [7]. First, rigid VTS typically relies on a fixed camera-surface geometry and stable illumination path to preserve photometric consistency for tactile reconstruction. Introducing pneumatic deformation would perturb this geometry and violate the optical conditions required for accurate structured-light sensing. Second, actuator-induced surface displacement can move the sensing layer away from the nominal camera focal plane, leading to blur and degraded fine-grained texture or color cues. In contrast, MoiréSkin encodes deformation into low-frequency, large-scale Moiré fringes that remain visually salient across different actuation states, including under moderate defocus, as shown in Fig. 7 (a), (b). This property makes MoiréSkin naturally compatible with integrated pneumatic actuation, enabling a sensor-actuator VTS that supports both tactile perception and active compliance control.

5 SOFTWARE DESIGN

Software runs on a control and computation unit connected to the camera and actuator. We first describe actuation-state estimation, followed by tactile sensing.

5.1 Sensing Layer Actuation State Estimation

Our calibration infers pneumatic state from Moiré geometry: a one-time barometer-instrumented material model maps fringe rotation to pressure, while a barometer-free visual zero-state procedure re-anchors it after environmental changes. For closed-loop control, MoiréSkin estimates whether the layer is flat, inflated, or deflated from $\Delta B = B_{in} - B_{air}$, where B_{in} and B_{air} denote chamber and ambient pressure. Instead of two pressure sensors, it infers ΔB from fringe rotation θ_M , preserving compactness and sensitivity.

5.1.1 Fringe-to-Pressure Modeling. Following Sec. 3.2.2, we infer ΔB from Moiré fringe rotation θ_M . Without external contact, the zero-state is $\Delta B = 0$ ($B_{in} = B_{air}$), where the flat gratings have equal pitch $p_{upper}^{(0)} = p_{lower}^{(0)}$ and reference orientation $\theta_M^{(0)}$ (Fig. 3(a)). We estimate ΔB from $\Delta\theta_M = \theta_M - \theta_M^{(0)}$. Preliminary results show a monotonic but nonlinear mapping. Inflation and deflation deform the thickness-separated gratings differently, yielding $R_p = p_{upper}/p_{lower} \neq 1$ and a nonlinear optical response through Eq. 2. Strain hardening further reduces marginal deformation as ΔB increases [55, 56]. The mapping therefore depends on membrane thickness, compliance, and stretchability and requires empirical calibration.

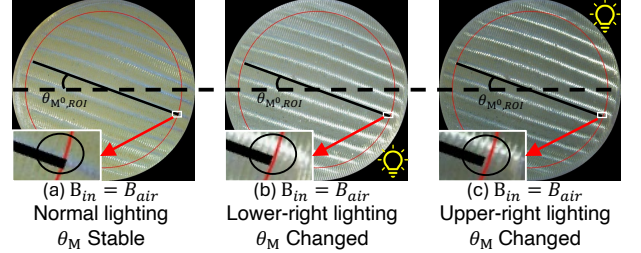


Figure 6: Fringe stability under different lighting

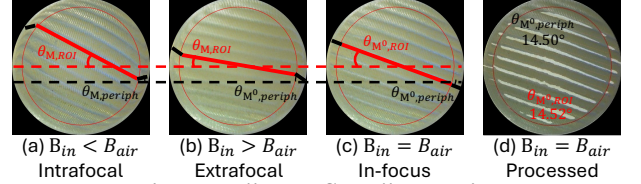


Figure 7: "Zero-State" detection

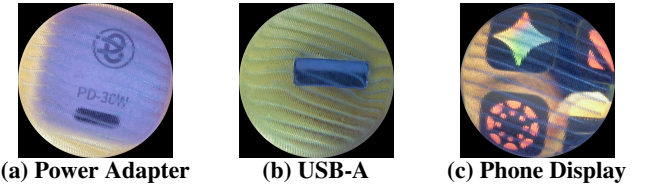


Figure 8: "See-through" functionality.

Material Calibration. Two commercial MEMS barometers (Adafruit MPRLS) temporarily measure chamber and ambient pressure; both are removed afterward. This characterization is performed once per membrane configuration. Cold starts use only the visual zero-state procedure in Sec. 5.1.2, without barometers, and full recalibration is required only after membrane or material changes. Using pump and vacuum control, we stabilize 100 randomly sampled targets across -11 to $+12$ hPa and record θ_M . We identify $\theta_M^{(0)}$ at $\Delta B = 0$, compute $\Delta\theta_M$, and fit it to ΔB . For the current material configuration, a third-order polynomial provides a good approximation:

$$\widehat{\Delta B}(\Delta\theta) = -0.0108\Delta\theta^3 + -0.1748\Delta\theta \quad (3)$$

The calibrated function enables barometer-free pressure inference during normal operation.

5.1.2 "Zero-State" Calibration. Atmospheric pressure and illumination shift the reference orientation $\theta_M^{(0)}$ across environments. Even at one location, B_{air} may change by 10–20 hPa per day [57], comparable to MoiréSkin's -11 to $+12$ hPa actuation range and sufficient to passively inflate or deflate the layer. Moreover, the transparent cover enabling the see-through observations in Fig. 8 (Sec. 4.2) exposes the fringes to moving shadows and changes in light position or intensity. Even at constant pressure, illumination can shift the apparent orientation by several degrees (Fig. 6(b,c)). These changes motivate re-estimating the zero-state at each cold start.

At the zero-state ($\Delta B = 0$), the sensing layer is flat. Fig. 7 divides its fringes into a central ROI and a peripheral region. Because the rigid frame constrains the outer annulus [58], the peripheral and central regions develop different strain profiles during inflation or deflation, so their orientations

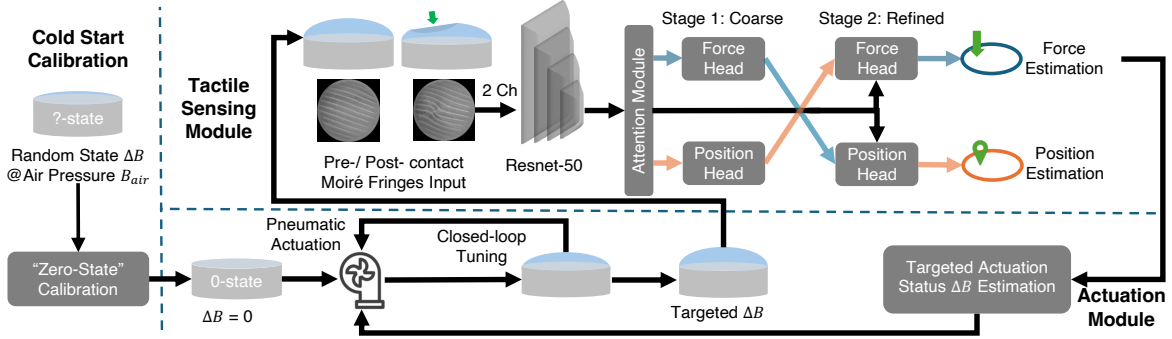


Figure 9: MoiréSkin software pipeline.

evolve differently. Only at the zero-state do the black peripheral and red central fringes form one continuous line with $\theta_{M,Periph} = \theta_{M,ROI}$ (Fig. 7(c)); this equality detects the flat state.

We convert the image to the LAB L-channel, use CLAHE [59] to enhance local contrast, and apply Gaussian blur to suppress high-frequency texture while preserving the low-frequency fringes. An adaptive Gaussian-weighted threshold isolates locally dark regions, and contour-based principal component analysis estimates the dominant orientation in each region. Area-weighted averaging improves the peripheral estimate against edge effects and non-uniform illumination (Fig. 7(d)).

The controller monitors $\Delta\theta_{M,ROI-Periph} = \theta_{M,ROI} - \theta_{M,Periph}$. When the layer is concave, $\theta_{M,ROI} < \theta_{M,Periph}$ indicates $B_{in} < B_{air}$, so the pump inflates it (Fig. 7(a)). For a convex layer, the inequalities reverse and the pump deflates the chamber. A PID controller modulates pump duty cycle from this error and reaches the zero-state within 2 s.

After reaching the zero-state, the system records its fringe orientation as $\theta_M^{(0)}$ and estimates subsequent $\widehat{\Delta B}$ using Eq. 3. Repeating this cold-start procedure compensates for environmental drift caused by atmospheric-pressure and ambient-light changes.

5.2 Tactile Sensing Module Design

The sensing novelty is a pre-/post-contact, two-stage network that conditions force on estimated contact position and position on estimated force, explicitly accounting for the coupled mechanics of an inflatable surface. The tactile sensing module leverages contact-induced distortions in the Moiré fringes to infer fine-grained contact position and normal force in real time. However, accurate estimation is challenging because the optical response depends not only on the contact itself, but also on the instantaneous actuation state of the sensing layer. First, during real-world operation, both the contact shape and the applied force are unknown, producing local Moiré distortions that are more complex than those observed during calibration (Sec. 5.1). Second, the same force applied at different locations can generate markedly different Moiré distortion patterns because the sensing surface is not always planar. Under different actuation states, the surface may become curved, so the same load applied near the apex can produce a different response from the same load applied near

the periphery. Third, even for the same contact position and force, the observed Moiré distortion may vary across actuation states. Pneumatic actuation changes the sensing-layer geometry, causing both the apparent contact position and the resulting deformation pattern to shift and become defocused, as shown in Fig. 7. To address these challenges, we develop a neural-network-based framework for simultaneous contact position and normal force estimation.

5.2.1 Model Input and Output. Pre-/Post- Contact Inputs. The model takes two images as input: one captured before contact and the other after contact. For each raw image, we first apply a circular mask to isolate the sensing ROI from the full MoiréSkin cell. We then convert each image from BGR to the CIELAB L-channel to decouple Moiré fringe distortions from chromatic variations. The resulting grayscale image is further normalized so that the mean brightness within the ROI is mapped to 128 (examples shown in Fig. 12). This step reduces sensitivity to variations in LED intensity and ambient illumination.

Model Output. Following tactile sensing literature [60], the model predicts two quantities: (1) contact position, defined as the center of the contact region; and (2) normal contact force.

5.2.2 Backbone Neural Network. MoiréSkin adopts a modified ResNet-50 architecture [61] with a custom weight initialization strategy to adapt pre-trained features to single-channel geometric fringe distortions. To bridge the domain gap between natural images and tactile patterns, we initialize the first convolutional layer by summing pre-trained RGB kernels and replicating them across all input channels, preserving edge and orientation detectors while removing color-specific bias. A Convolutional Block Attention Module (CBAM) [62] is integrated before global pooling to emphasize localized fringe distortions and suppress background interference from the transparent sensing layer.

5.2.3 Two-Stage Tactile Sensing Refinement. As discussed above, the inflatable geometry and varying actuation states of the sensing layer lead to position-dependent force responses and can also bias contact position. These nonlinear effects are difficult to capture with a single-stage predictor. We therefore design a two-stage, physics-informed refinement framework, as shown in Fig. 9

Stage 1: Coarse-Grained Estimation. The latent features extracted by the ResNet-50 backbone are first processed by two regression heads: a coarse position head and a coarse

force head. These heads generate initial estimation of the contact position and normal force magnitude.

Stage 2: Fine-Grained Refinement. We then use the Stage 1 outputs as physics-informed conditioning variables for refinement. For force refinement, the coarse position estimate is concatenated with the backbone latent features and fed into a refined force head. This allows the model to adapt force estimation to the location-dependent stiffness of the membrane. For position refinement, the coarse force estimate is concatenated with the same latent features and passed to a refined position head. This stage compensates for position shifts caused by actuation-induced geometric deformation. Together, these two refinement paths improve both force and position accuracy by modeling their mutual dependence.

5.2.4 Training Data Augmentation. During model training, we apply a physics-informed data augmentation pipeline to improve the model robustness.

Actuation-State Augmentation. To simulate the geometric variability introduced by pneumatic actuation, we apply random perspective transformation with distortion scale 0.15 and probability 0.5. This augmentation approximates the stretching and fringe-orientation changes that arise across different actuation states, helping the network learn features that are less sensitive to global shape variation.

Ambient-Light Augmentation. To improve robustness to changing optical conditions, we apply color jitter with brightness 0.4 and contrast 0.4 at probability 0.8. This models illumination and contrast variation caused by ambient light changes, shifting light paths, and partial light occlusion.

Low-Light Robustness: We further inject additive Gaussian noise with standard deviation 0.01 at probability 0.5 to simulate sensor noise under low-light conditions. This improves robustness when fringe observations are degraded by insufficient illumination or camera noise.

6 SETUP AND IMPLEMENTATION

6.1 Hardware Setup

Fig. 1 shows the hardware setup; the sensing and actuation layers follow Sec. 4.1. A 5 V pump with vacuum control is driven by an Arduino-controlled L298N driver for inflation, deflation, and PWM regulation, with an external supply providing its high current. An Arducam IMX291 RGB camera (2 MP, 1920×1080) with a 160° lens monitors the layer. A WS2812B 8-bit RGB ring (455–485 nm) excites the lower-layer fluorescent dopants, while a frosted base film diffuses illumination and reduces hotspots. A PETG housing rigidly aligns the chamber, camera, and illumination. The camera and Arduino connect to a MacBook Pro.

Fabrication Cost and Replicability. We separate reusable tooling from the recurring per-cell bill. Patterned molds, the plasma cleaner, ultrasonicator, spin coater, vacuum desiccator/oven, and microscope are amortized across cells. Per-cell inputs comprise the 9:1:1 LSR formulation (including the 2–3 mL dispensed in the single-layer casting step), fluorescent additive, TPU cover, printed PETG housing, RGB camera,

LED ring, pump, and driver. The MacBook Pro and external supply are evaluation infrastructure rather than integrated cell components. No specialized cleanroom is required.

Repeatability follows fixed process parameters: 30 W mold plasma treatment for 2 min, 5 min mixing, 2 min ultrasonication, vacuum degassing until visible bubbles disappear, and a 65°C vacuum bake for 10–15 min. Each dual-grating film is inspected microscopically before hardware integration. A film passes this gate when both patterns transfer without delamination, incomplete regions, or residual bubbles; rejected films are remade before adding the camera and pneumatic components. We define fabrication yield at this gate as accepted dual-grating films divided by total attempted films. This staged inspection localizes process loss to membrane fabrication, while reusable molds and fixed material ratios support repeatability across cells.

6.2 Software and Inference Pipeline

The software stack is implemented in PyTorch and OpenCV, and all training and inference are performed on an NVIDIA RTX 6000 GPU. The tactile network has 28.23M parameters and a 107.69 MB FP32 footprint. The latency reported in Sec. 7.4 establishes its desktop-GPU performance and provides a concrete baseline for onboard optimization. We train two models for evaluation: (1) core tactile sensing and (2) downstream shear-force application.

Tactile Sensing Model. For contact position and normal force estimation, we optimize the model using AdamW with a weight decay of 1×10^{-4} and an initial learning rate of 1×10^{-4} . We use a ReduceLROnPlateau scheduler (factor 0.5, patience 7), a batch size of 32, and early stopping with a patience of 50 over a maximum of 500 epochs. To improve robustness in the highly non-linear sensing space, we use Smooth L1 loss, automatic mixed precision (AMP), and gradient clipping with a maximum norm of 1.0. Its evaluation results are reported in Sec. 7.2.2. We also directly reuse this model, without additional fine-tuning, for ultra-soft material stiffness classification in Sec. 7.3.1.

Shear-Force Fine-Tuning. The shear-force refinement model is initialized from the pre-trained tactile sensing model and fine-tuned using AdamW with an initial learning rate of 5×10^{-5} . We use a batch size of 64, gradient clipping with a maximum norm of 5.0, and early stopping with a patience of 20 over a maximum of 200 epochs. This model is used for the shear-force downstream application evaluation in Sec. 7.3.2.

7 EVALUATION

We evaluate MoiréSkin from four perspectives. First, we assess the accuracy of sensing-layer actuation state estimation (Sec. 5.1). Second, we evaluate tactile sensing accuracy, including contact position and normal force estimation (Sec. 5.2). Third, we present two case studies to demonstrate the benefits of MoiréSkin’s integrated sensing and actuation: (1) soft material stiffness classification (Sec. 7.3.1) and

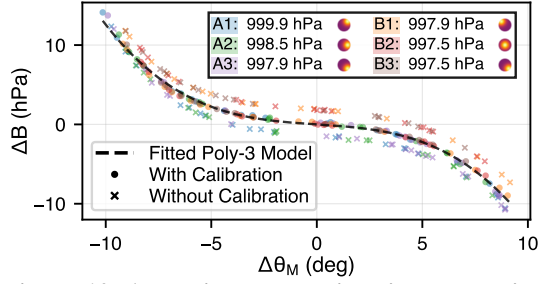


Figure 10: Actuation state estimation evaluation.

(2) mass estimation via shear force (Sec. 7.3.2). Fourth, we evaluate the system latency and model size.

7.1 Actuation State Estimation Accuracy

We evaluate whether our material calibration and “zero-state” calibration (Sec. 5.1) can be accurate to estimate the actuation state, i.e., ΔB .

7.1.1 Evaluation Setup & Data Collection. To establish ground truth for ΔB , we instrumented MoiréSkin with two MEMS barometers as discussed in Sec. 5.1.1. We apply a recursive EMA filter ($\alpha = 0.1$) to the raw pressure stream to suppress electronic jitter and high-frequency noise, while regulating target states via a PID-controlled air pump. To ensure pneumatic equilibrium, data points are logged only when the signal standard deviation remains below 0.2 hPa, followed by a mandatory 2-second stabilization window. Final ground-truth values are derived from an average of 10 consecutive readings to minimize measurement uncertainty of barometer. After 100 repetitions to establish the baseline pressure-fringe relationship between ΔB and $\Delta\theta$ (Eq. 3). We collect 120 test points across six sessions. All sessions followed a 2-hour sampling interval to capture long-term temporal stability. We introduce joint variations in ambient atmospheric pressure B_{air} and illumination to evaluate long-term temporal and environmental stability.

7.1.2 Experimental Results. Fig. 10 shows that without “zero-state” calibration (“x” marker), the regression function degrades significantly under varying atmospheric conditions with R^2 values dropping as low as 0.772. The “zero-state” calibration makes the regression function consistently restore the R^2 to above 0.994 and achieves a reduction in RMSE of up to 85%. The ΔB MAE reduces from 1.771 hPa to 0.151 hPa even during high-drift scenarios (Sessions B). This confirms that the “zero-state” calibration guarantees stable actuation state sensing without the need for external pressure sensors, enabling closed-loop sensing and actuation.

7.2 Tactile Sensing Accuracy

We evaluate contact-position and normal-force sensing at two scales, followed by an ablation of the design in Sec. 5.2.

7.2.1 Evaluation Setup & Data Collection. A sensor cell on a 1,000 g-capacity, 0.01 g-resolution scale is indented by an iDraw 2.0 XY plotter; pre-/post-contact frames form each input. The general-force setting uses a 15 mm flat probe (Figs. 11(a) and 12(a)), randomized $\Delta B \in [-11, 12]$ hPa, and forces up to 4 N. The tiny-force setting uses a 0.72 mm

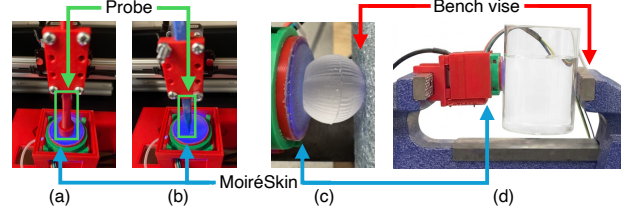


Figure 11: Experimental setups used in § 7.2 & § 7.3

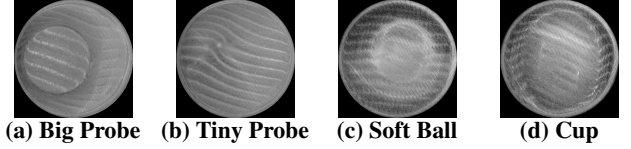


Figure 12: Processed Internal Sensor Views.

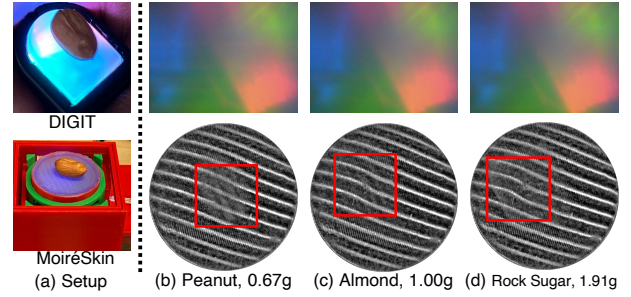


Figure 13: Qualitative Comparison: (a) Setup. (b–d) Sensor imaging with DIGIT and MoiréSkin

pointed probe at zero-state to maximize sensitivity below 400 mN (Figs. 11(b) and 12(b)). Each setting has 4,000 training/validation and 1,000 test samples over 256/400 Poisson-disk positions, respectively. Per position, we use 10–13 randomized indentation depths and re-home the plotter every 50 probes to limit drift. Because test positions are disjoint, this protocol directly evaluates spatial generalization across both controlled probe geometries. Separate models are trained for the two settings.

Separate models use 5-fold cross-validation; we report force and position MAE, RMSE, and R^2 on the test sets.

7.2.2 Experimental Results. In the general-force setting (Fig. 14), force response remains highly linear across 4 N, and 94.9% of force errors are below 0.1 N despite optical shifts across randomized actuation states. Position MAE rises slightly above 3 N because of focal shifts, but overall accuracy remains high. In the tiny-force setting (Fig. 15), errors remain stable across load levels and force-error bars are typically below 15 mN even near 400 mN. Position error is submillimeter throughout the sensing area, with a slight rise near the frame (10.4–12.0 mm radius) likely caused by boundary constraints. Overall, MoiréSkin reaches 0.094 mm position and 3.78 mN normal-force MAE, supporting gram-scale perception beyond existing VTS [34].

7.2.3 Ablation Study. To quantify the impact of our key design choices, we compare the proposed model against two baselines: (1) “I-Ch, W/o Aug.”, which uses only a single-channel post-contact image without data augmentation, and (2) “I-Ch, W/ Aug.”, which adds data augmentation to the single-channel input. As shown in Tab. 2, our full design

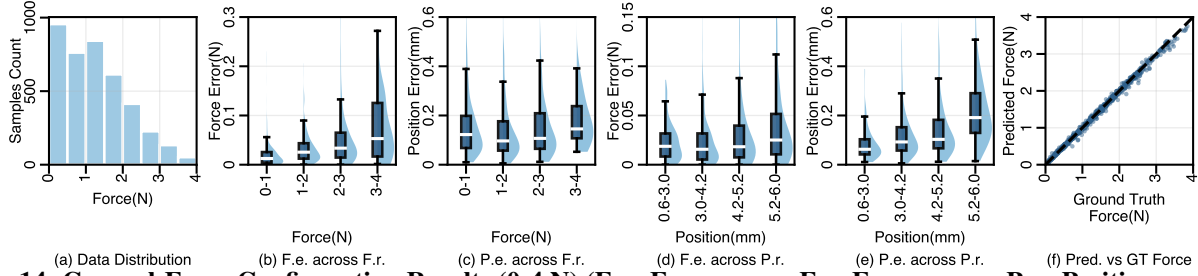


Figure 14: General-Force Configuration Results (0-4 N) (F.e.: Force error; F.r.: Force range; P.e.: Position error; P.r.: Position range. Position range uses an equal-area binning strategy, where each bin represents an identical spatial extent.)

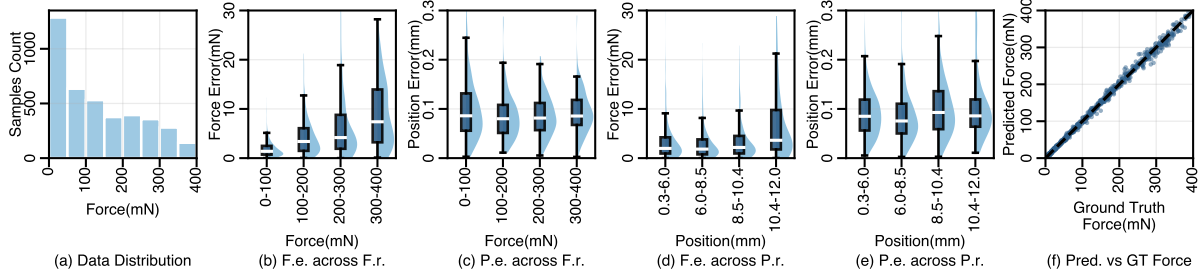


Figure 15: Tiny-Force Configuration Results (0-400 mN)

Table 2: Ablation Study: Quantifying the performance gap in contact position and force estimation by 5-fold cross validation. Red arrows (↓) indicate the relative performance degradation compared to our framework.

Configuration	Pos. MAE [mm]	Degradation	Force MAE [N]	Degradation
<i>General-force Configuration (0 – 4 N)</i>				
1-Ch, W/o Aug.	0.1749 ± 0.0236	↓ 70.0%	0.0318 ± 0.0018	↓ 28.7%
1-Ch, W/ Aug.	0.2149 ± 0.0204	↓ 108.8%	0.0319 ± 0.0023	↓ 29.1%
2-Ch, W/ Aug.	0.1029 ± 0.0032	Proposed	0.0247 ± 0.0005	Proposed
<i>Tiny-force Configuration ($\Delta B = 0, 0 - 0.4$ N)</i>				
1-Ch, W/o Aug.	0.1188 ± 0.0102	↓ 19.5%	0.0068 ± 0.0013	↓ 44.7%
1-Ch, W/ Aug.	0.1148 ± 0.0056	↓ 15.5%	0.0052 ± 0.0004	↓ 10.6%
2-Ch, W/ Aug.	0.0994 ± 0.0075	Proposed	0.0047 ± 0.0004	Proposed

consistently improves sensing accuracy and reduces measurement variance in both the general-force and tiny-force configurations. In the general-force configuration, it reduces position and force MAE by 70.0% and 28.7%, respectively, compared to “1-Ch, W/o Aug.”. In the tiny-force configuration, it reduces position and force MAE by 19.5% and 44.7%, respectively, relative to “1-Ch, W/o Aug.”. Overall, these results show that our model design improves generalization across varying actuation states and lighting conditions.

The gain over both one-channel baselines suggests that the pre-contact image supplies a physical reference that complements augmentation by separating contact deformation from actuation-dependent background shifts.

7.2.4 Comparison with DIGIT [19]. Fig. 13 qualitatively compares MoiréSkin with GelSight DIGIT [19] by placing gram-scale objects on both sensors. Under such challenging load scenarios, DIGIT exhibits indistinguishable change, whereas MoiréSkin produces clear localized Moiré fringe distortions.

Together with the 3.78 mN MAE in Fig. 15, this comparison shows how the measured tiny-force sensitivity translates into visible gram-scale contact cues.

7.3 Integrated Sensing & Actuation Studies

Next, two fixed-vise studies isolate MoiréSkin’s integrated sensing–actuation behavior: (1) material-stiffness classification and (2) dynamic mass estimation from shear force. Together, they evaluate manipulation-relevant sensing under repeatable grasp conditions. For both cases, MoiréSkin is mounted on one jaw of a bench vise, with the object grasped at a fixed vise position to isolate tactile changes induced by actuation and other object mechanics, as shown in Fig. 11 (c) and (d), with the internal sensor views shown in Fig. 12 (c) and (d) respectively. We use a higher actuation state, $\Delta B = 41.25$ hPa, to create the stiffer contact interface needed for stable grasping. This setting extends beyond the standard training range (−11 to +12 hPa); fine-tuning adapts the pretrained tactile model to this operating point.

7.3.1 Case Study 1: Soft Material Stiffness Classification. This study evaluates whether MoiréSkin can discriminate among seven ultra-soft materials that differ only in stiffness, with Young’s moduli ranging from 8.73 kPa to 21.72 kPa, roughly covering tofu-like to skin-like softness, as shown in (Fig. 16). To mimic humans’ active probing, each ball is grasped once at the fixed vise setting and imaged first at $\Delta B = 0$, then again after inflating the chamber to $\Delta B = +41.25$ hPa. This produces a two-state tactile image pair for each trial, and we collect 60 sessions in total.

For classification, we directly use the 2048-dimensional latent features extracted by our pre-trained general-force tactile model, and applied Linear Discriminant Analysis (LDA) to the features. Fig. 17 shows that MoiréSkin achieves an exclusive 10-shot accuracy of 99.7%, proving that the pre-trained backbone model can be used directly for this study. The complementary 5-shot setting reaches 92.7% accuracy across the seven evaluated stiffness classes in the 5-shot



Figure 16: Seven Balls Soft Material Test Set

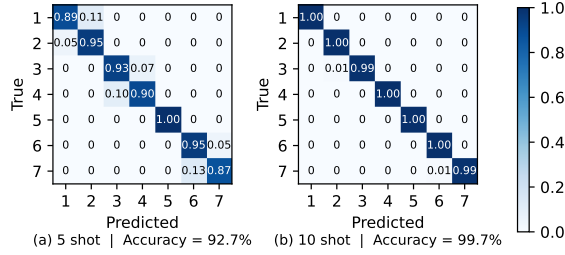


Figure 17: Stiffness Classification Confusion Matrices

regime. This high accuracy highlights its potential for industrial sorting and medical robotics requiring soft-material identification.

7.3.2 Case Study 2: Dynamic Mass Estimation via Shear Force. This case study evaluates MoiréSkin’s ability to sense shear force during manipulation by tracking the changing mass of water in a grasped cup, where added liquid produces a corresponding increase in gravity-driven shear load. The same cup is grasped at a fixed vise position and constant actuation state ΔB while water is gradually added from 0 to approximately 300 mL, and Moiré images are recorded at 10 Hz, as shown in Fig. 11(d). We collect six sessions, using one unseen session for testing and the remaining five for cross-validation.

To adapt the tactile sensing model to this task, we reuse the pre-trained tactile sensing backbone from the general-force configuration. We then fine-tune the ResNet-50 backbone so that it captures the subtle fringe bending induced by the downward pull of the liquid mass and train a new shear-force regression head from scratch to estimate the mass. Fig. 18 shows that the estimation mass error remains stable across the full mass range. The system achieves a MAE of 4.1462 ± 0.5994 g and a RMSE of 4.7517 ± 0.5353 g, respectively. Even at larger masses (e.g., 250–350 g), where the induced shear stress is higher, the error does not increase substantially. These results indicate that the Moiré fringe distortions encode sufficiently rich mechanical information to support tactile inference beyond normal force estimation alone. **These results establish the fixed-vise prototype as a dynamic tactile scale and quantify the mass-change signal available for pressure regulation under changing loads.**

Scope and Future Work. The current evaluation establishes a clean systems baseline for sensing accuracy and sensing-actuation coupling under controlled probing and grasping. The next evaluation stage will scale the sensing area; add geometry-diverse and fragile/deformable contacts; broaden object classification; and integrate closed-loop tasks on a robot end effector. Dedicated pneumatic-cycling and shear-fatigue tests will complement the repeated-use evidence in Sec. 4.1.

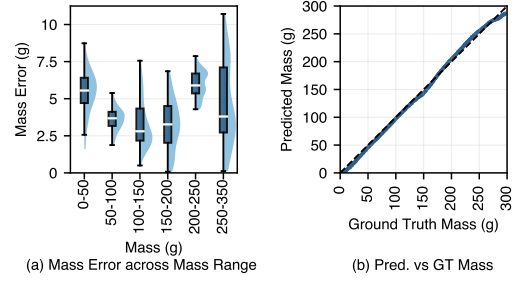


Figure 18: Dynamic Mass Estimation

7.4 System Evaluation

System Latency. (1) *Pneumatic actuation and calibration.* Across the tested initial pressure states, both pneumatic actuation and zero-state calibration converge within approximately 2 s; their response is governed by the pump, membrane mechanics, and pressure-control loop rather than neural inference. (2) *Visual tactile readout.* The tactile model takes 0.756 ± 0.085 ms per sample on an NVIDIA RTX 6000 GPU. Because the camera supports up to 120 fps, acquisition bounds visual readout at approximately 8.3 ms per frame. Thus, contact estimation operates at a faster time scale than pneumatic compliance regulation, which is intended for slower pressure adjustments rather than millisecond contact transients.

Model Size and Memory Requirements. The tactile model contains 28.23M parameters and occupies 107.69 MB in FP32. On the evaluated desktop GPU, inference is comfortably faster than camera acquisition and pneumatic regulation. These measurements define a concrete optimization target for Jetson-class deployment.

The current external supply powers the camera, computation unit, and pneumatic module; integrating power and onboard computation is the next step toward a self-contained platform.

8 CONCLUSION

We presented MoiréSkin, an inflatable visuo-tactile skin that embeds dual-sided sub-millimeter gratings in a soft membrane to unify ultra-sensitive tactile sensing and active compliance regulation. By transforming microscopic contact-induced strain into low-frequency, high-contrast Moiré fringes, we repurpose geometric interference to visualize microscopic contact deformations and internal actuation states. Extensive evaluations demonstrate that it achieves state-of-the-art tactile sensing precision, while maintaining “see-through” functionality. MoiréSkin points to a new direction for achieving ultra-sensitive sensor-actuators for fine-grained robotic manipulation and tactile dynamics.

Together, these results establish a systems foundation for highly sensitive, actively compliant tactile interfaces based on Moiré. The next phase will extend this foundation to larger sensing areas, geometry-diverse contacts, onboard computation, and closed-loop robotic manipulation.

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